



DPUJoystickConsole

Virtual Mockup which is operated via a joystick.

1. Inputs:

<i>JoystickX_In</i>	X axis of joystick nomalized to [-1,+1]
<i>JoystickA_In</i>	Input for accelerator pedal position
<i>JoystickB_In</i>	Input for brake pedal position
<i>JoystickC_In</i>	Input for clutch pedal position
<i>Button4_In</i>	Input for the buttons
<i>GearUp_In</i>	Button to upshift
<i>GearDown_In</i>	Button to downshift
<i>v</i>	Speed of vehicle
<i>HShift</i>	Flag for H-shift control
<i>ButtonGearR</i>	Button for gear "R" (H-shift)
<i>ButtonGear[1..6]</i>	Buttons for gear 1..6 (H-shift)

2. Dynamic parameters:

<i>SteeringLinMax</i>	Maximum limit of linear range of <i>JoystickX_In</i>
<i>SteeringLinMin</i>	Minimum limit of linear range of <i>JoystickX_In</i>
<i>SteeringLinGain</i>	Resulting overall gain
<i>SteeringLinGain1</i>	Gain factor 1 of linear range
<i>SteeringLinGain2</i>	Gain factor 2 of linear range
<i>VSteeringLinGain1</i>	Speed-dependent gain limit in m/s
<i>VSteeringLinGain2</i>	Speed-dependent gain limit in m/s
<i>SteeringPolExp</i>	Potenz of distortion function
<i>SteeringWheelMax</i>	Maximum value of steering wheel angle
<i>SteeringWheelMin</i>	Minimum value of steering wheel angle
<i>AcceleratorPedalGain</i>	Gain factor for the accelerator pedal
<i>BrakePedalGain</i>	Gain factor for the brake pedal
<i>JoystickAMin</i>	Minimum value for the accelerator pedal position
<i>JoystickAMax</i>	Maximum value for the accelerator pedal position
<i>JoystickBMin</i>	Minimum value for the brake pedal position
<i>JoystickBMax</i>	Maximum value for the brake pedal position
<i>JoystickCMin</i>	Minimum value for the clutch pedal position
<i>JoystickCMax</i>	Maximum value for the clutch pedal position

3. Outputs:

<i>Ignition</i>	Ignition on
<i>Starter</i>	Starter on
<i>HandBrake</i>	Hand brake
<i>Gear</i>	Selected gear in range of [-1,+5]



<i>SteeringWheel</i>	Steering wheel angle in Rad
<i>AcceleratorPedal</i>	Accelerator pedal position in range of [0,+1]
<i>BrakePedal</i>	Brake pedal position in range of [0,+10]
<i>ClutchPedal</i>	Clutch pedal position in range of [0,+1]
<i>Button4_Out</i>	Outputs for the buttons

4. Functional principle:

The DPUJoystickConsole represents an interface of a virtual mockup to the joystick. It is connected between vehicle dynamics software and a DPU querying joystick in-put (e.g. DPUDX8Joystick, DPUKeyDX8Joystick).

From the x axis of the joystick, the steering angle is calculated as follows:

$$SteeringWheel = SteeringWheelExp + SteeringLinGain \cdot JoystickX$$

In the area [*SteeringLinMin*,*SteeringLinMax*], the steering wheel angle is linear to the joystick movement. The linear amplification *SteeringLinGain* can be changed by the parameters *SteeringLinGain1*,*SteeringLinGain2* and the parameters of *vSteeringLinGain1* and *vSteeringLinGain2*.

$$\begin{aligned}
 &SteeringLinGain \\
 &= \frac{(SteeringLinGain2 - SteeringLinGain1) \cdot (v - vSteeringLinGain1)}{(vSteeringLinGain2 - vSteeringLinGain1)} \\
 &+ SteeringLinGain1
 \end{aligned}$$

Within the linear area parameter, *SteeringWheelAlpha* is set to 0.

$$SteeringWheelExp = \begin{cases} SteeringWheelAlpha & | \quad JoystickX > SteeringLinMax \\ 0 & | \quad \text{else} \\ SteeringWheelAlpha & | \quad JoystickX < SteeringLinMin \end{cases}$$

In the areas outside of the linear limits, the parameter *SteeringWheelAlpha* is calculated as follows:

$$\begin{aligned}
 &SteeringWheelAlpha \\
 &= ((SteeringPolExp - 1) \cdot |JoystickX| - SteeringLinMxx)^{SteeringPolExp}
 \end{aligned}$$

Output Gear reacts to the positive edges of the input signals *GearUp_In* and *GearDown_In*.

Value	Gear
-1	Reverse gear
0	Neutral / parking
1	Gear 1
2	Gear 2
3	Gear 3
4	Gear 4
5	Gear 5